

ABDULLAH HASHMAT

AI/ML Researcher • LLMs & Agentic Systems • NLP & Computational Biology
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EDUCATION

BS Computer Science | Lahore University of Management Sciences (LUMS)

Aug 2021 – May 2025

Relevant Coursework: CS6304 Advanced Machine Learning, CS5302 Speech Processing with Generative AI, CS6303 Large Language Models, CS331 Artificial Intelligence, CS487 Cloud Development, CS437 Deep Learning

TECHNICAL SKILLS

Programming: Python, C++, C, C#, Java, SQL, Haskell

AI/ML: PyTorch, Transformers, LLMs, Agentic AI, RAG, NLP, Computer Vision, Fine-tuning, Knowledge Distillation

Cloud & MLOps: AWS (Lambda, ECS, DynamoDB, S3), Docker, Distributed Training, Model Deployment, Serverless Architecture

Frameworks & Tools: Hugging Face, LangChain, RDKit, Git, Linux, React

EXPERIENCE

Research Fellow | Max Planck Institute for Software Systems (MPI-SWS) | Germany, KL

Apr 2026 – Present

Supervised by Dr. Abhilasha Ravichander

- Designed LLM-driven agentic AI systems for inverse molecular design, combining generative models with structured tool-calling (RDKit, docking, QSAR) for multi-objective optimization pipelines.
- Built closed-loop generation pipelines with iterative refinement and feedback driven reasoning for self evolving agentic systems, improving validity, diversity, and binding affinity across candidate molecules.

Researcher | Center for Speech and Language Technologies (CSaLT), LUMS | Lahore, PK (Remote)

Sep 2024 – Present

Supervised by Dr. Agha Ali Raza & Dr. Usman Naseem

- Developing self-evolving agentic frameworks with built in cultural awareness, enhancing LLM reasoning and tool calling performance while reducing Western centric alignment biases.
- Published PakBBQ, Pakistan's first locally contextualized QA bias benchmark (17,180 QA pairs across 8 socio-cultural categories), accepted at EMNLP 2025 Main Conference.
- Led development of Pak3H, a human-validated Urdu benchmark suite (28,000 samples) evaluating LLM alignment across helpfulness, harmlessness, and honesty, exposing significant cross lingual degradation.

Lead Researcher | German Research Center for Artificial Intelligence (DFKI) & RPTU | Germany, KL

Jul 2025 – Mar 2026

Supervised by Dr. Nabeel Asim

- Developed a multi-representation SMILES-based Molecular Property Prediction model using a multi granular transformer framework leveraging atom-, functional-group, and branch level representations.
- Pre trained BERT, RoBERTa, DeBERTa, and state space models on large scale SMILES corpora, applied multi teacher knowledge distillation (MLM + FitNet) with dynamic gating to compress into a unified student model, achieving state of the art performance across all 22 ADMET benchmark tasks.
- Built and processed a data pipeline processing a 500M+ compound dataset spanning 150+ molecular properties via large scale web scraping, ETL, and distributed GPU training clusters.

MLOps Engineer | Darcheeni (AI Healthcare Startup, LUMS) | Lahore, PK

May 2024 – May 2025

- Led backend development for an AI-driven clinical assistant integrating ASR, RL-based decision models, and LLM-driven response generation for real-time patient triage.
- Optimized speech recognition pipelines through data processing, model fine tuning, and LLM based post processing, improving transcription quality across diverse clinical settings.
- Designed and deployed serverless infrastructure on AWS (Lambda, DynamoDB, Auth0) with autoscaling for scalable, lowlatency model inference.

Primary Researcher | Center of Urban Informatics, Technology and Policy (CITI) | Lahore, PK

Sep 2024 – Mar 2025

Supervised by Dr. Muhammad Tahir

- Proposed a confidence-aware multi-teacher knowledge distillation framework to mitigate texture/shape inductive biases and improve OOD generalization in medical image classification.
- Ensembled shape- and texture-biased ViT and VGG16 teachers using entropy-based logit weighting; trained ViT-Small student via KL divergence with dynamic bias propagation.
- Achieved balanced cue reliance (shape 0.49 / texture 0.51) and outperformed single-teacher and naive KD baselines across ImageNet-1k, Stylized-ImageNet, and OOD benchmarks.

Sales Performance Intern | [Delivery Hero \(Foodpanda\)](#) | Lahore, PK

Jun 2024 – Aug 2024

- Conducted large-scale EDA on vendor onboarding and QC data to identify bottlenecks, reducing QC rejection rate by 15%.
- Developed an agentic Python workflow using Google APIs to automate daily data pulls, invoice generation, and vendor distribution via WhatsApp, reducing process time by 35%.

Data Science Intern | [Pakistan Television Network \(PTV\)](#) | Lahore, PK

Jun 2023 – Jul 2023

- Revamped ticker headline system using ML text similarity models, achieving a 20% reduction in headline redundancy.
- Built a predictive content analysis model that improved viewer engagement metrics by 10% through data-driven strategies.

SELECTED PROJECTS

AI-Driven Stock Insights Platform (AWS + Claude API) [Link](#)

Sep 2024 – Dec 2024

- Built and deployed full-stack real-time stock analysis platform using AWS Lambda, S3, ECS, and Neon Postgres; integrated Claude-based chatbot for natural language investment queries.
- Achieved <200ms latency with auto-scaling to millions of users via serverless architecture and ECS Fargate.

Bias Mitigation in Deep Learning Vision Models [Link](#)

Sep 2024 – Dec 2024

- Implemented multi-teacher single-student knowledge distillation (ViT + VGG16) to address inductive biases; achieved 12% bias reduction on cue-conflict datasets under distributional shift.

Agentic Data Science Chatbot [Link](#)

Jan 2024 – May 2024

- Built an agentic AI chatbot for automated EDA, statistical analysis, and data cleaning across 10+ formats; fine-tuned Llama 3 and deployed on Hugging Face via Gradio.

PUBLICATIONS & RESEARCH

PakBBQ: Culturally Adapted Bias Benchmark for QA — EMNLP 2025 Main Conference [Paper Link](#)

Pak3H: Evaluating the Cost of Cultural Mismatch in LLM Alignment — EMLP 2026 Main Conference [Link](#)

IRTIQA: Evolving in English, Evaluated in Urdu — Does Agent Self-Evolution Transfer Across Languages? — Under ARR Review (2026) [Link](#)

TEACHING ASSISTANTSHIPS

AI600 Machine Learning (Masters in AI Program) | LUMS

Jan 2025 – May 2025

CS5302 Foundations of Generative AI | LUMS

Jan 2025 – May 2025

CS535 Machine Learning | LUMS

Sep 2024 – Dec 2024

CS100 Computational Problem Solving | LUMS

Sep 2023 – Dec 2023

LEADERSHIP & ACTIVITIES

TechStars Startup Weekend Lahore — Convenor

Jul 2022 – Mar 2023

- Organized Startup Weekend Lahore 2023 with National Incubation Center & Google for Startups; managed 40+ startups, 150+ attendees, team of 60, and secured 800K+ PKR in sponsorships.

SPADES (PSIFI) — Assistant Director, Socials & Operations

Sept 2022 – May 2023

- Led organisation of PSIFI's flagship event, coordinating social functions and concerts for 1,400+ attendees